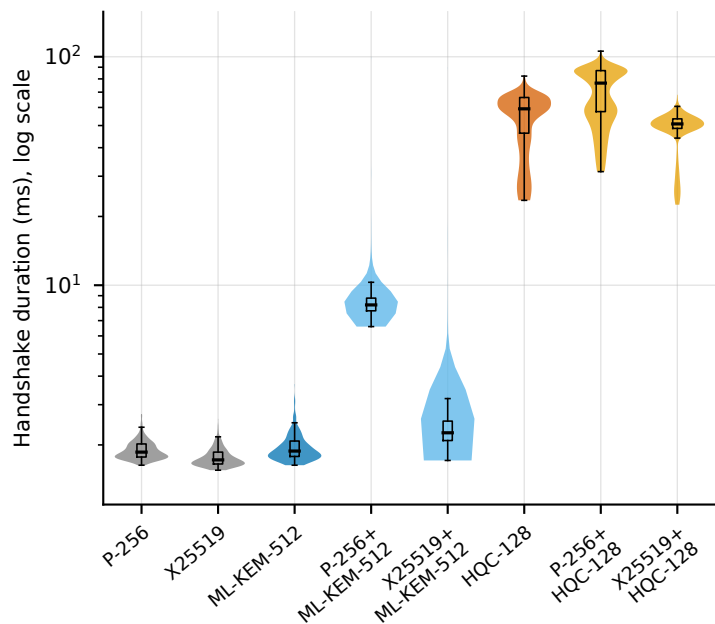


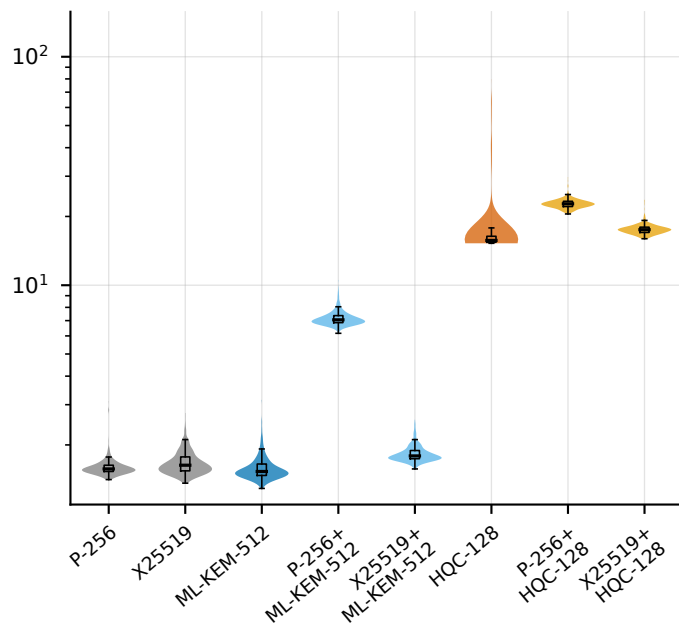
(a)

ed25519 (classical)



(b)

mlds44 (post-quantum)



(c)

Classical KEX
 ML-KEM (pure)
 ML-KEM (hybrid)
 HQC (pure)
 HQC (hybrid)

KEM	ed25519 Mean (ms)	ed25519 CV (%)	ed25519 Max (ms)	mlds44 Mean (ms)	mlds44 CV (%)	mlds44 Max (ms)
P-256	1.91	9.8	2.73	1.60	10.1	3.11
X25519	1.77	9.5	2.60	1.69	12.3	2.76
ML-KEM-512	1.97	14.5	3.69	1.60	12.6	3.15
P-256+ML-KEM-512	8.68	50.3	99.5	7.18	7.5	10.6
X25519+ML-KEM-512	2.66	152.5	90.0	1.84	8.5	2.58
HQC-128	55.0	25.2	82.3	18.4	50.8	79.8
P-256+HQC-128	70.1	25.3	105.8	22.8	4.5	29.7
X25519+HQC-128	49.1	17.3	68.2	17.6	5.0	23.7